

LOHIT

Backend Engineer

Hisar, Haryana, India | Open to relocation | +91 951-849-2408 | 11undefeated22@gmail.com
linkedin.com/in/lohit-undefeated | github.com/Undeclared22 | lohit-portfolio-drab.vercel.app

SUMMARY

Backend engineer, B.Tech 2026. Go, TypeScript, Node.js, PostgreSQL. I build systems that survive failure and prove it: a durable-execution engine whose crash recovery is verified by deterministic simulation, an LLM incident-investigation backend on pgvector, and a brokerage ledger where money is BigInt and balances are derived, never mutated.

EXPERIENCE

Freelance Backend Engineer — Brokerage Client (contract)

2026 – Present | Next.js 16, TypeScript, PostgreSQL 17, Prisma 7, Zod

- Sole engineer on a trading client portal: 55 routes over an 80-model Postgres schema spanning identity, KYC, ledger, markets, copy trading and analytics.
- Modelled money as BigInt minor units on an append-only ledger with derived balances; composite foreign keys let Postgres reject a ledger leg in a currency the account cannot hold.
- Centralised authorization server-side: one data-access layer for all user-scoped reads, KYC and funding gates as server guards, TOTP step-up auth, four-eyes approval on admin withdrawals.
- Streamed live quotes and candles over Server-Sent Events behind mockable provider seams; wrote 28 invariant checks that each fail if a past defect returns.

Smartworks — Gurugram, India

Software Development Intern | Jun 2025 – Sep 2025

- Deployed and customized Twenty (open-source CRM) from local evaluation into a self-hosted, team-facing internal product.
- Diagnosed KPI-counting and ticket-assignment defects by tracing wrong results back through MongoDB aggregation pipelines to the failing stage.
- Built ElasticSearch-backed search across snag and user records, replacing direct collection scans.
- Shipped FCM push notifications for WorkCtrl, a React Native app live on the App Store and Play Store, and integrated Formbricks feedback capture.

PROJECTS

dex — Durable Execution Engine | [GitHub](#)

Go, PostgreSQL, deterministic simulation testing

- Built a Temporal-style durable execution engine in Go: workflows resume at the exact step they stopped after a crash, replaying an append-only Postgres event log instead of persisting mutable state.
- Drove every node, clock and scheduler from one seeded PRNG in a single-goroutine simulator — whole-cluster runs reproducible from a seed, with automatic shrinking; one triage cut 44 injected faults to the 2 that caused the bug.
- Implemented shard leases with fencing tokens to stop split-brain writes, transfer and timer queues with ack levels, and a determinism linter that rejects non-replayable calls in workflow code.
- Verified recovery by killing a live process mid-execution against Postgres; shipped dexctl and an embedded dashboard for event histories and live workflow state.

Forge — AI Incident-Investigation Backend | [Live Demo](#) | [GitHub](#)

Node.js, Fastify 5, PostgreSQL/pgvector, Redis/BullMQ, Groq, Gemini, Railway

- Ingests alerts from Grafana, PagerDuty, Sentry, CloudWatch and Opsgenie and runs autonomous LLM root-cause investigations, embedding each incident into pgvector so prior failures inform new ones.
- Hardened ingestion against duplicate and racing webhooks with incident fingerprinting and atomic Postgres upserts; field redaction, at-rest encryption, RBAC-scoped multi-tenant isolation.
- Built a provider-agnostic LLM seam (Groq generation, Gemini embeddings) with two-tier routing and quota-aware rate limiting, so one provider's exhaustion degrades the pipeline instead of breaking it.

EDUCATION

Guru Jambheshwar University of Science & Technology — Hisar, India

B.Tech, Electronics & Communication Engineering | Graduated May 2026

Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Object-Oriented Programming, System Design

TECHNICAL SKILLS

Languages: Go, TypeScript, JavaScript (ES6+), Python, SQL, Java, C/C++

Backend: Node.js, Fastify, Express.js, REST APIs, event sourcing, background workers, BullMQ, Socket.io, Server-Sent Events, rate limiting, JWT auth, RBAC, multi-tenant isolation

Databases: PostgreSQL, pgvector, MongoDB, Redis, SQLite, ElasticSearch, Prisma, Drizzle ORM

Infrastructure: Docker, GitHub Actions, CI/CD, Linux, Railway, Vercel

AI: LLM APIs (Groq, Gemini), embeddings, vector search, prompt engineering

Frontend: React, Next.js, React Native, Tailwind CSS, Radix UI

ADDITIONAL

Smart India Hackathon college-level qualifier — led an alumni-association platform build for a government engineering college. Maintained the official Training & Placement Cell website, GJUS&T.